



LG-TriCapsNet: A lightweight graph capsule framework with nearest neighbor graphs for multi-disease EEG classification

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Highlights

- A novel LG-TriCapsNet model integrates Nearest Neighbor Graphs (NNG) and Capsule Networks.
- The model effectively captures spatial-temporal dependencies in EEG for disease classification.
- Achieves 98.34% accuracy, surpassing state-of-the-art methods in multi-disease classification.
- Demonstrates robustness to noise and missing EEG channels in real-world clinical scenarios.
- Optimized for real-time deployment with low memory (3.8M parameters) and fast inference (47.9ms).

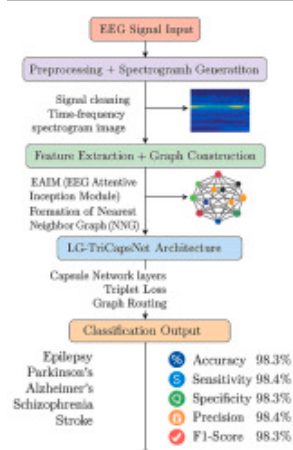
Abstract

EEG signal classification for neurological disorders is a very critical task in the healthcare field, demanding accuracy and efficiency. Due to the diversity of these disorders and the complexity of the EEG signals, the task of diagnosing these disorders is very challenging. Traditional methods

usually cannot capture the intricate relationships between temporal and spatial features, which are inherent in EEG data. To overcome this challenge, the proposed Lightweight Graph Triplet Capsule Networks combined with Nearest Neighbor Graphs effectively leverage both temporal and spatial information in EEG signals. The graph-based representation enhances the spread of information across the network, which thus performs efficient feature extraction and good classification performance. By integrating NNG, the model dynamically captures spatial-temporal dependencies between EEG channels, significantly improving feature discrimination across neurological conditions. The novelty of the approach lies in the combination of graph structures and capsule networks, which particularly makes it suited for handling the complexities of EEG data. LG-TriCapsNet is performing quite well as the F1 score is 98.32%, accuracy is 98.34%, sensitivity is 98.30%, and specificity is 98.40%, and it also outperforms the current state-of-the-art models, which points to the fact that this framework is really effective for the classification of such a variety of neurological disorders. LG-TriCapsNet, by providing a computationally efficient, real-time solution, holds great potential for advancing clinical decision-making and patient care by offering a robust tool for automated neurological disease detection and diagnosis. At

<https://github.com/jainshraddha12/LG-TriCapsNet> ↗, the source code will be available to the public.

Graphical abstract



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Introduction

The primary goal of our study is to develop a new technique for brain diseases detection, aligning it with the World Health Organization's plan in 2022. This involves using EEG data despite their complexity low signal strength and high background noise issues as valuable sources that can provide significant details about brain activity. Our strategy that primarily aims at recognizing and treating common neurological disorders, then fostering efforts towards enhancing quality care with better data collection within WHO initiatives plays a pivotal role in these important fields [1]. The information on brain function and brain disorders is critical as EEG signals are representative of the electrical activity produced by the brain itself [2]. One of the interesting methods in identifying

those neurological disorders (like stroke, epileptic seizures, schizophrenia, parkinson) involves the use of spectrogram images. These images show the changes in a visual manner from the time-frequency domain for EEG signals [3]. This technique we used as part of a multi-disease neurological disorder categorization problem addressed in our most recent study; we aim at achieving higher levels of precision and efficiency when it comes to disease categorization task. This application used for deep learning-based feature extraction and classification methods, integrated with the spectrogram images [4].

Recent developments in deep learning and graph-based approaches have greatly enhanced the capability of EEG signal classification. Of these, the most commonly adopted approaches in feature extraction with regards to EEG signals in neurological disease detection are CNNs, RNNs [5], and LSTM networks [6]. Such methods have the ability to learn complex, hierarchical patterns in data but generally tend to be inefficient at capturing the dependencies both in temporal and spatial in EEG signals. Such dependencies are critical to proper diagnosis. Recently, graph-based models such as Graph Convolutional Networks (GCNs) and Graph Neural Networks (GNNs) have emerged as powerful tools for capturing spatial relationships and topological structures in EEG signals, thereby significantly improving the classification accuracy for diseases like Alzheimer's, schizophrenia, and epilepsy [7].

Recent developments in EEG-based diagnosis of neurological disorders have led to significant improvements in classification accuracy and disease detection. Such of the developments by Yu et al. [8] include the use of supervised network-based fuzzy learning, as presented in integrating fuzzy logic and supervised learning for the improved classification of EEG signals. This does account for the uncertainty in the EEG data, consequently improving the detection of Alzheimer's disease in terms of its capacity to capture complex, imprecise relationships. Adebisi et al. proposed the rising technique of EEG-based brain functional network analysis for differential identification of dementia-related disorders [9]. This method allows the differentiation of the types of dementia and tracking the onset: based on the analysis of functional connectivity. Besides, the recent developments in graph-based deep learning techniques in medical diagnostics by Ahmedt-Aristizabal et al. [10] have shown promising capabilities to capture complex relationships in EEG data, leading to better diagnosis of neurological disorders. Finally, Yan et al. [11] reviewed how graph theory combined with both EEG and MRI data leads to the ability to diagnose neurological disorders by applying the strengths of both data types for more precise disease detection. These innovations present the growing importance of deep learning and graph-based approaches in the diagnosis and analysis of neurological diseases, especially in terms of EEG signal classification.

Recent advancements in Capsule Networks (CapsNet) and graph-based models have been promising in the analysis of EEG signals for neurological disorder classification. However, these approaches fail to capture both spatial and temporal dependencies between EEG channels effectively. Though CapsNet has been successful in learning hierarchical features from EEG signals, it cannot represent dynamic interactions between brain regions over time. In this work, we put forward a new approach called Lightweight Graph Triplet Capsule Network, or LG-TriCapsNet for short. It's built by integrating NNG with Capsule Networks, enabling the model to capture both spatial and temporal relationships within EEG data. Our approach addresses major limitations to previous methods by modeling intricate brain connectivity patterns effectively and providing

superior classification performance in multi-disease classification tasks such as epilepsy, Alzheimer's, or Parkinson's disease.

Apart from these advanced capabilities of feature extraction, the interpretability of graph-based features that arise from the EEG signal is also explored here. Such a feature plays a vital role in unveiling the structural and functional relationships within brain regions. Our graph-based features are compared with traditional EEG analysis methods, such as functional brain networks and modulation of spectral power, which have been widely used in EEG analysis, including studies on acupuncture modulation of brain activity. These comparisons help us to assess the value of the proposed method in capturing brain connectivity more comprehensively. Triplet network design holds the promise of extracting features that are class-separable; there are difficulties in optimizing its performance for classification. Although training for class-separable features is naturally sensitive, there is still a gap in efficiently converting metric loss into concrete classification results [12]. Finding the best classification architecture is made more difficult by inconsistent classification performances throughout training epochs, resulting in the need for a careful examination of feature–classifier combinations. It becomes challenging to identify the root cause of under-performance in the classifier, learns features, or both [13].

We propose LG-TriCapsNet, a multi-disease neurological classification system that integrates Nearest Neighbor Graph (NNG) and Lightweight Graph Triplet Capsule Networks. To extract useful features from complicated neurological data, LG-TriCapsNet utilizes capsule networks and graph-based representation learning [14]. Its main innovation is how it uses Nearest Neighbor Graph (NNG) to extract features from neurological datasets effectively and capture underlying structural links. LG-TriCapsNet also utilized triplet capsule networks to learn hierarchical features and spatial correlations. It reduced constraint on achievable end performance and offers a way to speed up the optimization process [15]. This work focuses on reducing dependency on labelled data by proposing the LG-TriCapsNet architecture. Combining nearest neighbor graph, our approach aims to fill the gap between the scarcity of labelled data and the need for robust and generalizable models in medical time series research. We present comparison research within a challenging patient-independent environment over three baselines from five different datasets.

We introduce a new multi-disease neurological classification framework called LG-TriCapsNet, a novel hybrid model which is a fusion of Nearest Neighbor Graphs (NNG) and Lightweight Graph Triplet Capsule Networks (CapsNet). The proposed model exploits spatiotemporal dependencies of EEG data and maintains hierarchical relations among features. With the integration of graph-based representation, LG-TriCapsNet effectively models local and global structural patterns within EEG signals and mitigates the shortfalls of existing approaches. In LG-TriCapsNet, NNG facilitates efficient extraction of structural and functional correlations among EEG channels, while triplet capsule networks facilitate feature discriminability for enhanced classification performance. The combination addresses the prevalent issue of sparse labelled EEG data by limiting dependence on large datasets and enhancing generalization across diverse neurological disorders. Hierarchical feature extraction through capsule networks guarantees strong representation learning, which is essential to address the complexity of neurological EEG datasets.

EEG signals have spatiotemporal correlations such that brain areas dynamically interact over time. Nonetheless, conventional models such as CNNs and RNNs fail to capture spatial dependencies and

hierarchical relations. CapsNet is great at learning hierarchical spatial features but fails to incorporate an explicit mechanism for modeling functional connectivity between EEG channels. To fill this deficit, we incorporate Nearest Neighbor Graphs (NNG) that capture functional connectivity between EEG channels and support feature propagation to achieve effective disease classification. We treat EEG channels as graph nodes and their interactions as weighted edges and utilize NNG to tighten feature relations and improve classification.

The proposed framework for the identification of multiple neurological disorders from EEG signals is summarized in Fig. 1. The process of ensuring an in-depth evaluation of the model's quality to distinguish between neurological disorders is depicted in the diagram in multiple steps.

In this paper, our main work includes:

Our contributions include a complete methodology for the EEG-based classification of neurological diseases:

1. We propose Nearest Neighbor Sparse Inverse Covariance (NNSIC) to represent EEG signals as a graph structure, which can capture the spatiotemporal relationship among EEG channels for the model.
2. We introduce LG-TriCapsNet, an integration of Capsule Networks (CapsNet) with Nearest Neighbor Graphs (NNG). Such an integration enables the model to retain hierarchical relationships of features and at the same time use graph structures to enhance feature extraction.
3. We utilize Graph Routing Algorithms in place of standard pooling methods in order to maintain spatial relationships to better provide for classification accuracy. Adding Triplet Loss Optimization boosts the feature separability further to increase the ability of the model to separate and classify distinct neurological diseases.
4. Our model has 98.34% accuracy (F1: 98.32%), outperforming state-of-the-art models such as 1D-CapsNet (97.44%) and AlexNet+CNN (95.79%). LG-TriCapsNet is tested on several public EEG datasets and demonstrates excellent generalization ability, even when tested with noisy and missing EEG signals.

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Section snippets

Related works

Recent advancements in EEG classification have explored Capsule Networks (CapsNet) for hierarchical feature learning. However, standard CapsNet models primarily function as local feature extractors and do not model spatiotemporal interconnections among brain regions. Meanwhile, Graph Neural Networks (GNNs) and functional connectivity methods capture global brain correlations but often lack fine-grained feature extraction required for disease classification.

To address these limitations, we ...

Methodology

Integration of Nearest Neighbor Graph (NNG) with Capsule Networks (CapsNet) is the novel contribution for EEG-based classification of neurological disorders. NNG allows the model to capture spatial-temporal dependencies between EEG channels that have not been fully captured by previous methods. Although traditional CapsNet learns hierarchical feature representations, it generally doesn't consider the complex interactions of brain regions over time. We can better model the connectivity between ...

Datasets descriptions

This work employs five publicly released EEG datasets covering various neurological disorders to assess the performance and generalizability of LG-TriCapsNet. The datasets are Epilepsy (Ep), Parkinson's Disease (Pd), Alzheimer's Disease (Ad), Schizophrenia (Sz), and Stroke (St), providing a diverse range of brain activity patterns that shows in Table 1. The experiment employs EEG datasets from multiple publicly accessible sources to provide diversity in patient populations, recording ...

Advantages

The inclusion of Nearest Neighbor Graphs inside Capsule Networks inside the LG-TriCapsNet model improves upon capturing spatial and temporal dependency abilities by 98.34%. As such, this model tends to outscore several of the previously proposed algorithms, thereby offering better generalizability over neurologically based disorders and as such is very reliable for classification with multiple types of diseases, thereby offering key interpretability. The researchers and clinicians may be in a ...

Conclusion and future work

In our LG-TriCapsNet, NNG was developed as our final model for representing multi-disease neurological classification and this is truly a giant step. With graph-based data representation, successful integration was realized with the Capsule Network in our proposed model, attaining an accuracy of 98.34% with the test samples. This result demonstrates that graph-oriented approaches with capsule networks are very effective at improving the potential to apply them for diagnostic use in medical ...

CRedit authorship contribution statement

Shraddha Jain: Writing – original draft, Validation, Software, Methodology, Conceptualization.

Rajeev Srivastava: Writing – review & editing, Visualization, Supervision, Software, Resources, Formal analysis, Conceptualization. ...

Consent to participate

Not applicable. ...

Consent for publication

Not applicable. ...

Data and code availability

Datasets used in this study are publicly available and accessible <https://archive.ics.uci.edu/dataset/388/epileptic+seizure+recognition> ↗ <https://openneuro.org/datasets/ds002778/versions/1.0.2> ↗ <https://openneuro.org/datasets/ds004504/versions/1.0.6> ↗ <https://repod.icm.edu.pl/dataset.xhtml?persistentId=doi:10.18150/repod.0107441> ↗ https://figshare.com/articles/dataset/EEG_datasets_of_stroke_patients/21679035 ↗ The code used in this paper are available at <https://github.com/jainshraddha12/LG-TriCapsNet> ↗ for ...

Ethics statement

This study was conducted in accordance with the ethical principles outlined in the Declaration of Statement. All EEG data used in this research were obtained from publicly available datasets, ensuring that appropriate ethical guidelines for data collection and privacy were adhered to. No personal identifiers were used, and all data were anonymized prior to analysis. Additionally, all research involving human subjects was approved by the relevant ethics committees, and informed consent was ...

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Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Shraddha Jain reports was provided by Indian Institute of Technology (BHU) Varanasi. SHRADDHA JAIN reports a relationship with Indian Institute of Technology (BHU) Varanasi that includes: non-financial support. The authors declare no conflicts of interest related to the research or the publication of this manuscript. Additionally, none of the authors have served ...

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